

Lucas Picard

B.S./M.S. Mechanical + B.S. Robotics Engineering | Mechatronics · CAD · Embedded Systems · Prototyping

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Portfolio

SUMMARY

Mechanical engineering student at WPI (3.75 GPA, Dean's List) on a combined B.S./M.S. in Mechanical Engineering with a second major in Robotics Engineering. I design and build real hardware, including embedded electronics, robotic systems, and CAD assemblies, and I direct AI to write the software and automation around them. Dual US/German citizen, bilingual in English and German.

EXPERIENCE

Microboard Processing - *Engineering Intern, Document & Controls*
Seymour, CT - contract electronics manufacturer (ITAR-regulated).

Jun - Aug 2026

- Reviewed customer board requirements and wrote the matching process flows, supporting about \$12 million in quoted work.
- Automated documentation work engineering did by hand, with an industrial engineer, cutting manual effort worth an estimated \$12,000 a year.
- Wrote the standard process-flow documents for SMT and PCB assembly, then presented new assemblies to the engineering subteams and project managers.
- Covered core Document & Controls Manager duties for the last weeks of the internship, handling ITAR-controlled documentation to the company rules.

Picard Studio - *Studio Assistant & Operations*

Feb 2020 - Present

- Built and maintained KPI dashboards tracking MRR, ARPU, retention, churn, and acquisition.
- Automated monthly metrics reporting through the Kajabi API and GitHub Actions, sending data straight into Google Sheets.
- Delivered freelance CAD work - designing, prototyping, and invoicing a wall-mount equipment cradle.

EDUCATION

Worcester Polytechnic Institute - *B.S./M.S. Mechanical Engineering + B.S. Robotics Engineering*

Expected 2029

Worcester, MA - 21 AP transfer credits. GPA: 3.75 / 4.00. Dean's List (Fall 2025, Spring 2026).

Completed coursework: Statics, Dynamics, Thermodynamics, Manufacturing Science & CNC Machining, Intro to Robotics, Intro to ECE, Calculus III-IV, Differential Equations, Linear Algebra.

Pomperaug High School - *Southbury, CT*

2021 - 2025

ENGINEERING PROJECTS

SCARA Robot Arm - *Personal mechatronics build*

2026 - Present

- Turned a salvaged \$150 Anet A8 3D printer into a teleoperated 4-axis SCARA arm for about \$120 in new parts, with my own 3D-printed arm segments and GT2 pulleys up to 100 teeth.
- An Xbox controller drives the arm at 25 Hz, with a Python host solving the inverse kinematics live and sending G-code to Marlin.
- Integrated AS5600 absolute encoders on a three-bus I²C design for switch-free homing; base, elbow, and Z are live, with wrist and gripper next.

PenGPT AI Smart Pen - *Embedded hardware / PCB design*

Mar 2026 - Present

- Co-designing the hardware with a teammate for a pen that turns handwriting on ordinary paper into text, tracking its own motion with two IMUs and a magnetometer around a Sophgo SG2002 Linux processor. Boards are targeted for November 2026.
- Verified the 268-part KiCad schematic pin by pin against manufacturer datasheets, catching three faults ERC missed. The 4-layer bring-up board is placed and routing is next.

CT Used-Car Scraper - *Python automation tool*

2026

- Directed AI to build a Python + Playwright system that scrapes 36 Connecticut dealer sites every 6 hours on GitHub Actions for \$0/month, tracks price history in SQLite, scores deals, and sends Discord alerts. It found the car I bought.

FRC Robotics - *Team Captain & Lead Driver · 1000+ hours*

2022 - 2025

- Captain for two of three seasons. Set direction and worked with the subteam leads, including marketing and outreach. Built the preseason training system of lessons and assignments, then moved the team onto project-management software for roles and tasks.
- Introduced CAD to a team that had none and was its only CAD student, taking the robot from taped-together scrap to fully modeled designs, and ran four iterations with subsystem prototypes in one season, up from a single un-prototyped design.
- Replaced the team's WAGO wiring with JST and bullet connectors, soldering every motor and controller over one February vacation, which ended the disconnections that had been costing matches.
- Started the team's presentations to the district Board of Education, schools and community groups. Worked 60-80 hour weeks through build season alongside a full course load, and as primary driver earned the team's first New England District Championship qualification in 10 years and its first CT State Championship.

TECHNICAL SKILLS

CAD & Prototyping: SolidWorks, Onshape, GD&T, design for manufacturability, 3D printing, manual milling, laser cutting, CAM

Electronics: KiCad schematic & PCB design, soldering, embedded systems (ESP32 / Arduino), closed-loop control, sensors & encoders

Software & Data: Java (semi-proficient), read Python, and direct AI to build and ship the rest: automation scripts, web scrapers, REST API integrations, GitHub Actions, and robot control hosts. Technical documentation, ITAR-aware data handling.

AWARDS & LANGUAGES

Awards: Dean's List (WPI, Fall 2025 and Spring 2026); Honor Roll (Pomperaug High School, all years).

Languages: English (native), German (Seal of Bilingualism, dual US/German citizen).